1. Consider the following subset of the real numbers

$$S = [-18, 1] \cup (3, \infty)$$

(a) Is the set open, closed, both open and closed, neither open nor closed? Justify your answer. (5)

(b) Determine $S^c = \mathbb{R} \setminus S$ and the set of interior points of S^c. (5)

[**10**]

Solution:

A very similar exercise was in the first tutorial sheet.

(a) The set S is not open, because for example the point 1 is not an interior point, any ϵ-neighbourhood of 1 contains at least a point $x > 1$, hence it is not contained in S. The set S is also not closed because it does not contain the point 3, which is a limit point of S because every deleted neighbouhood of 3 contains a point of S and a point in S^c.

(b) The complement of S is given by

$$S^c = (-\infty, -18) \cup (1, 3]$$

and the set of its interior points is

$$(S^c)^0 = (-\infty, -18) \cup (1, 3)$$

♣

2. Prove, using the ϵ and δ definition of a limit, that

$$\lim_{x \to 0} x \cos\left(\frac{1}{x^3}\right) = 0.$$

[**10**]

Solution:

In class I did a very similar exercise, I computed $\lim_{x\to 0} x \sin\left(\frac{1}{x}\right) = 0$ **with the** ϵ **and** δ **def of limit**
We remark the function $x\cos(\frac{1}{x^3})$ is defined in a deleted neighbourhood of 0 and that that for all $x \neq 0$

$$|x \cos\left(\frac{1}{x^3}\right)| \leq |x|.$$

Hence for all $\epsilon > 0$ there exists $\delta = \epsilon$ such that if $0 < |x| < \delta$, then

$$|x \cos\left(\frac{1}{x^3}\right)| \leq |x| < \delta = \epsilon.$$

♣

3. (a) Define what it means for a function $f : [a, b] \to \mathbb{R}$ to be Lipschitz continuous. (5)

   (b) Obtain the smallest Lipschitz constant for the following function on its domain of definition

   $$f : [0, 4] \to \mathbb{R}, \quad f(x) = (x + 1)e^{-x}.$$

   (5)

Solution: [**10**]

(a) **seen - definition central to course**
A function $f : [a, b] \to \mathbb{R}$ is *Lipschitz continuous* if $\exists\ L > 0$, s.t.

$$|f(x) - f(y)| \leq L|x - y|, \qquad \forall\ x, y \in [a, b].$$

The constant L is called a *Lipschitz constant* for f.

(b) **seen similar in homework sheets and in lectures**
We have that $f'(x) = e^{-x} - (x + 1)e^{-x} = -xe^{-x}$ and $f''(x) = xe^{-x} - e^{-x} = (x - 1)e^{-x}$. So, f' has a local turning point at $x = 1 \in [0, 4]$. Thus

$$L = \max\{|f'(0)|, |f'(1)|, |f'(4)|\} = \max\{0, e^{-1}, 4e^{-4}\} = \frac{1}{e}$$

♣

4. (a) Let $n \in \mathbb{N}$, $f_n : [a, b] \to \mathbb{R}$ and $f : [a, b] \to \mathbb{R}$. Suppose $f_n(x) \to f(x)\ \forall x \in [a, b]$. State suitable conditions for which it follows that

   $$\lim_{n\to\infty} \int_a^b f_n(x)\mathrm{d}x = \int_a^b f(x)\mathrm{d}x\,?$$

   (5)

   (b) Hence or otherwise, evaluate the limit

   $$\lim_{n\to\infty} \int_1^2 e^{-nx^2}\mathrm{d}x.$$

   (5)

Solution: [**10**]

(a) **seen - theorem central to course**
It is sufficient that the sequence of functions, $\{f_n\}_{n=1}^{\infty}$, is a sequence of Riemann integrable functions defined on a finite interval $[a, b]$; and f_n converges uniformly to f as $n \to \infty$.

(b) **seen similar in lectures**
Since $f_n(x) = e^{-nx^2} \leq e^{-n}$ on $[1, 2]$, the sequence of functions, $\{f_n\}_{n=1}^{\infty}$, converges uniformly to 0 on $[1, 2]$. Hence the limit and integral can be exchanged, and so the limit is 0.

♣

5. (a) Define what it means for a function $f : \mathbb{R}^n \to \mathbb{R}^m$ to be differentiable at $p \in \mathbb{R}^n$. (5)

(b) Fix $b \in \mathbb{R}^n$ and let $f : \mathbb{R}^n \to \mathbb{R}$ be given by

$$f(x) = \langle x, b \rangle,$$

where $\langle a, b \rangle$ denotes the inner product of a and b.
Show that f is differentiable at each $p \in \mathbb{R}^n$. (5)

Solution: [**10**]

(a) **seen - theorem central to course**
The function $f : \mathbb{R}^n \to \mathbb{R}^m$ is differentiable at $p \in \mathbb{R}^n$ if and only if $\exists\ A \in L(\mathbb{R}^n, \mathbb{R}^m)$ such that if $g(x) = A(x - p) + f(p)$, then

$$\lim_{x \to p} \frac{\|f(x) - g(x)\|}{\|x - p\|} = 0$$

where $A \in L(\mathbb{R}^2, \mathbb{R})$ is a linear map $\mathbb{R}^n \to \mathbb{R}^m$. We write

$$A := Df(p)$$

and call $Df(p)$ the *differential of the map* f at point p.

(b) **seen similar in homework sheets**
Consider the map $f : \mathbb{R}^n \to \mathbb{R}$ given by $f(x) = \langle x, b \rangle$. To see that f is differentiable at each $p \in \mathbb{R}^n$, with $Df(p)[h] = \langle h, b \rangle$, we note that

$$f(x) - Df(p)(x - p) - f(p) = \langle x, b \rangle - \langle x - p, b \rangle - \langle p, b \rangle = 0$$

by the linearity of the inner product. Therefore, we have

$$\lim_{x \to p} \frac{\|f(x) - g(x)\|}{\|x - p\|} = 0.$$

♣

6. (a) Let a, b be real numbers. State, using ϵ and δ, what does it means for a function $f : [a, b] \to \mathbb{R}$ to be continuous from the right at a. (5)

   (b) Let $f(x)$ be the function

$$f(x) = \frac{\cos\left(\frac{1}{x}\right) e^{x^5}}{x^4 \sin x}.$$

   Is it true that there exists $M > 0$ such that $|f(x)| \leq M$ for all $1 \leq x \leq 2$? Justify your answer. (10)

   (c) Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function such that there exist $a, b, x_0 \in \mathbb{R}$ with $a \leq x_0 \leq b$ such that $f(x) \geq f(x_0)$ for all $x \in (-\infty, a) \cup (b, \infty)$. Prove that there exists $m \in \mathbb{R}$ such that $m = \min\{f(x) | x \in \mathbb{R}\}$. (10)

Solution: [**25**]

(a) **bookwork, definition central to course**
We say that $f : [a, b] \to \mathbb{R}$ is continuous from the right at a if

$$\lim_{x \to a^+} f(x) = f(a).$$

This means that f is defined in $[a, c)$ where $c \in \mathbb{R}$ and $\forall \epsilon > 0$ there exists $\delta > 0$ such that if $a \leq x < x + \delta$

$$|f(x) - f(a)| < \epsilon$$

(b) **unseen, but immediate application of a theorem proven in class, this theorem was proven in details, also two exercises in the problems sheets used that theorem. Moreover if the students prefer they can compute M explicitly, I will give them full mark in both cases.**
Yes, the function $f(x)$ is a continuous function on $[1, 2]$, because product of composition of continuous functions (remark that $0 < 1 < 2 < \pi$, so $\sin x \neq 0$, for all $x \in [1, 2]$). And we proved that a continuous function on a closed and bounded interval is bounded.

(c) **unseen**
The function f is continuous in $[a, b]$, which is closed and bounded, hence there exists $c \in [a, b]$ such that $f(c) \leq f(x)$ for all $x \in [a, b]$. In particular $f(c) \leq f(x_0)$ but $f(x_0) \leq f(x)$ for all $x \in (-\infty, a) \cup (b, \infty)$ by hypothesis, hence $f(c) \leq f(x)$ for all $x \in \mathbb{R}$. In other words $f(c) = m$.

♣

7. Suppose $a, b \in \mathbb{R}$ with $a < b$ and let $f : [a, b] \to \mathbb{R}$ be a bounded function. Let $P = (x_0, \ldots, x_k)$ be a partition of $[a, b]$.

   (a) Define the lower and upper Riemann sums $L(f; P)$ and $U(f; P)$. (5)

   (b) Define the lower and upper Riemann integrals of f

$$\underline{\int_a^b} f(x)\,\mathrm{d}x, \quad \overline{\int_a^b} f(x)\,\mathrm{d}x,$$

   and state what it means for a function f to be Riemann integrable. (10)

   (c) Let $f : [0, 1] \to \mathbb{R}$ be the function defined by $f(x) = x^2$, and let $\hat{P} = (0, \frac{1}{n}, \frac{2}{n}, \ldots, \frac{n-1}{n}, 1)$ be a partition. Use the previous parts of this question and partition $\hat{P}$, to prove that the function f is Riemann integrable over the interval $[0, 1]$.
   (You may use $\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$ without proving it.) (10)

Solution: [**25**]

(a) **seen - definition central to course**
Define

$$M_i = \sup_{x \in [x_{i-1}, x_i]} f(x)$$

$$m_i = \inf_{x \in [x_{i-1}, x_i]} f(x)$$

$$\Delta x_i = x_i - x_{i-1}.$$

Then, the lower Riemann sum of f with respect to partition P is

$$L(f; P) = \sum_{i=1}^{n} m_i \Delta x_i.$$

The upper Riemann sum of f with respect to partition P is

$$U(f; P) = \sum_{i=1}^{n} M_i \Delta x_i.$$

(b) **seen - definition central to course**
The lower Riemann integral of f over $[a,b]$ is

$$\underline{\int_a^b} f(x)\,\mathrm{d}x := \sup_P L(f,P).$$

The upper Riemann integral of f over $[a,b]$ is

$$\overline{\int_a^b} f(x)\,\mathrm{d}x := \inf_P U(f,P).$$

The function f is Riemann integrable if

$$\underline{\int_a^b} f(x)\,\mathrm{d}x = \overline{\int_a^b} f(x)\,\mathrm{d}x.$$

(c) **unseen**
We are given the partition $\hat{P} = (0, \frac{1}{n}, \frac{2}{n}, \ldots, \frac{n-1}{n}, 1)$. So, $x_i = \frac{i}{n}$ for $i = 0, 1, 2, \ldots, n$. Now, applying the formula for the Riemann sums, we have

$$\begin{aligned}
L(f;P) &= \sum_{i=1}^{n}(x_i - x_{i-1}) \inf_{x\in[x_{i-1},x_i]} f(x) \\
&= \sum_{i=1}^{n}\left(\frac{i}{n} - \frac{i-1}{n}\right) \inf_{x\in[\frac{i-1}{n},\frac{i}{n}]} x^2 \\
&= \frac{1}{n}\sum_{i=1}^{n}\left(\frac{i-1}{n}\right)^2 \\
&= \frac{1}{n^3}\frac{(n-1)n(2n-1)}{6} \\
&= \frac{(n-1)(2n-1)}{6n^2}
\end{aligned}$$

Also,

$$\begin{aligned}
U(f;P) &= \sum_{i=1}^{n}(x_i - x_{i-1}) \sup_{x\in[x_{i-1},x_i]} f(x) \\
&= \sum_{i=1}^{n}\left(\frac{i}{n} - \frac{i-1}{n}\right) \sup_{x\in[\frac{i-1}{n},\frac{i}{n}]} x^2 \\
&= \frac{1}{n}\sum_{i=1}^{n}\left(\frac{i}{n}\right)^2 \\
&= \frac{(n+1)(2n+1)}{6n^2}
\end{aligned}$$

Taking limits, we have

$$\lim_{n\to\infty} L(f;P) = \lim_{n\to\infty} \frac{(n-1)(2n-1)}{6n^2} = \frac{1}{3}$$

and

$$\lim_{n\to\infty} U(f;P) = \lim_{n\to\infty} \frac{(n+1)(2n+1)}{6n^2} = \frac{1}{3}.$$

Therefore, by definition, $f(x) = x^2$ is integrable on $[0,1]$.

♣